



Physics-Guided Hybrid Transformer Modelling for Microstructural Failure and Mechanical Response Prediction in High-Entropy Alloys

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Abstract: High-entropy alloys (HEAs) exhibit high mechanical strength, phase stability, and damage tolerance, making them promising materials for advanced structural applications. However, reliable prediction of their mechanical response and microstructural failure remains challenging because alloy composition, phase constitution, grain characteristics, thermodynamic descriptors, and deformation behaviour are strongly coupled. This study investigates a physics-guided hybrid modelling framework that integrates physically meaningful alloy descriptors with Transformer-based representation learning for failure and mechanical response prediction in HEAs. Mixing entropy, atomic size mismatch, valence electron concentration, phase stability parameters, and microstructural and mechanical descriptors were incorporated into the data-driven modelling architecture to establish physically informed relationships between material characteristics and failure behaviour. The developed framework was used to predict ultimate tensile strength (UTS), failure strain, stress–strain response, and microstructural failure probability. The results showed close agreement between the predicted and measured UTS and failure strain and demonstrated higher R^2 values than random forest, extreme gradient boosting (XGBoost), convolutional neural network (CNN), long short-term memory (LSTM), and standard Transformer models. The predicted stress–strain curves also reproduced the principal deformation behaviour observed in the experimental data. Failure-probability mapping and attention analysis identified the material and microstructural features associated with failure, while the ablation study showed that incorporating physics-guided information improved predictive accuracy, reliability, and interpretability. These findings demonstrate that the structured integration of physics-based descriptors and Transformer learning provides an effective hybrid modelling strategy for relating material constitution to mechanical response and failure. The proposed framework provides a physically informed and interpretable approach to failure prediction and data-assisted design of HEAs and offers a transferable modelling strategy for intelligent engineering analysis of complex material systems.

Keywords: High-entropy alloys; Hybrid modelling; Physics-guided learning; Transformer; Microstructural failure; Mechanical response prediction; Materials informatics

1 Introduction

High-entropy alloys (HEAs) constitute an important class of advanced engineering materials because of their high mechanical strength, thermal stability, corrosion resistance, and structural reliability under demanding operating conditions. Unlike conventional alloys dominated by one or two principal elements, HEAs contain multiple primary elements in near-equiatomic or comparable proportions, producing complex compositional and microstructural interactions that can generate distinctive mechanical behaviour [1]. Their broad compositional design space and unusual combinations of strength, ductility, phase stability, and environmental resistance have stimulated extensive research for applications in aerospace, energy, automotive engineering, and nuclear systems [2].

Despite these advantages, reliable prediction of the mechanical response and microstructural failure behaviour of HEAs remains difficult. Alloy composition, phase constitution, grain characteristics, thermodynamic stability, processing history, and deformation mechanisms interact across multiple scales, making conventional trial-and-error development both expensive and time-consuming. The large compositional space also limits the ability of experimental campaigns and high-fidelity simulations to systematically examine all relevant alloy configurations.

These challenges have motivated increasing use of machine learning (ML) to accelerate HEA property prediction, phase classification, and materials design [3].

Among the most established applications of ML to HEAs is phase prediction. Huang et al. [4] demonstrated that machine-learning models can use elemental and thermodynamic descriptors, including valence electron concentration and formation-related parameters, to distinguish among different HEA phase classes. Wen et al. [5] further showed that machine-learning-assisted alloy design benefits from domain-informed descriptors rather than relying exclusively on elemental composition. More recent studies have extended these approaches to broader HEA property prediction and alloy optimization, confirming the value of data-driven models for navigating high-dimensional composition–property spaces [6].

Although conventional ML models can provide effective predictions, they generally rely on predefined descriptors and statistical correlations. Their ability to represent complex interactions among heterogeneous material features can therefore be limited, particularly when compositional, thermodynamic, microstructural, and mechanical variables must be considered simultaneously. Transformer-based architectures provide an alternative because self-attention enables relationships among multiple input features to be learned directly. Kamnis and Delibasis [7] recently applied a Transformer-based model to HEA mechanical-property prediction and demonstrated that self-attention can capture complex elemental interactions relevant to ultimate tensile strength (UTS) and elongation. Attention-based architectures have also shown strong performance in more general materials-property prediction, where they provide both predictive capability and a degree of feature-level interpretability [8].

However, purely data-driven Transformer models do not inherently enforce physical consistency. Predictions can remain statistically accurate while violating known material trends or failing to generalize beyond the distribution represented in the training data. This limitation is particularly important for HEAs because phase stability, lattice distortion, thermodynamic effects, and microstructure-sensitive deformation mechanisms play central roles in determining mechanical behaviour. Consequently, integrating material-domain knowledge with modern representation learning offers a promising route toward more reliable and interpretable engineering models.

Against this background, this study develops a physics-guided hybrid Transformer modelling framework for microstructural failure and mechanical response prediction in HEAs. The framework combines physically meaningful descriptors with Transformer-based self-attention so that material-domain information and data-driven representation learning operate within a unified modelling structure. Mixing entropy, atomic size mismatch, valence electron concentration, phase-related descriptors, and experimentally reported microstructural variables are incorporated to establish physically informed relationships among alloy composition, microstructure, mechanical response, and failure behaviour.

The proposed framework addresses several complementary prediction tasks, including UTS, failure strain, stress–strain response, and microstructural failure probability. Attention-based feature analysis is employed to examine the relative importance of material descriptors, while comparative modelling and ablation analysis are used to evaluate the contribution of physics-guided information and Transformer-based representation learning.

The principal contributions of this study are as follows:

- A physics-guided hybrid Transformer modelling framework is developed for the integrated prediction of mechanical response and microstructural failure in HEAs;
- Thermodynamic, compositional, and microstructural descriptors are combined with Transformer-based representation learning to establish a structured connection between material-domain knowledge and data-driven modelling;
- The framework jointly addresses tensile strength, failure strain, stress–strain response, and failure probability rather than restricting prediction to a single material property;
- Attention-based interpretation is incorporated to identify the material features emphasized during the prediction process;
- Comparative and ablation analyses are used to assess the contribution of the individual modelling components to predictive performance, interpretability, and physical consistency.

Through this integration, the study treats physical knowledge and data-driven representation learning as complementary modelling components. The resulting framework is intended to improve not only predictive performance but also the physical coherence and interpretability of intelligent failure analysis, thereby supporting data-assisted engineering assessment and alloy design.

2 Literature Review

ML has become an increasingly important tool in materials informatics because it can approximate nonlinear relationships between composition, structure, processing conditions, and material properties. Jain and Bligaard [9] demonstrated the value of physically meaningful materials descriptors for large-scale property prediction and candidate-material screening. Such descriptor-based approaches provide an important foundation for data-driven

materials discovery, but their effectiveness depends strongly on the quality and physical relevance of the selected input representation.

For microstructure-sensitive materials, the integration of domain knowledge becomes especially important. Pokharel et al. [10] developed a physics-informed data-driven surrogate modelling approach for full-field three-dimensional microstructural and micromechanical evolution in polycrystalline materials. Their work showed that physical information can be used to evaluate and constrain data-driven predictions rather than treating the learning problem as purely statistical. This principle is directly relevant to HEAs, where compositional complexity and microstructural heterogeneity make physically unconstrained prediction particularly challenging.

Phase prediction provides a representative example of how physically meaningful descriptors can strengthen HEA modelling. Zhao et al. [11] used interpretable ML to construct descriptors for HEA phase prediction and demonstrated that combinations of physically motivated variables can distinguish among solid-solution, intermetallic, face-centered cubic (FCC), body-centered cubic (BCC), and amorphous configurations with high accuracy. Similarly, Jain et al. [12] compared multiple ML algorithms for HEA phase prediction and showed that model-based design can be combined with thermodynamic validation to support the development of new alloy compositions.

The broader development of high-throughput and data-driven approaches has further expanded the role of ML in HEA discovery. Lu et al. [13] reviewed the integration of high-throughput preparation, characterization, computation, and ML for establishing structure–property relationships in HEAs. Their analysis emphasized that efficient exploration of large compositional spaces requires not only predictive models but also reliable representations of the physical variables governing phase formation and mechanical performance.

Deep-learning architectures have extended this capability by learning higher-level representations directly from complex material inputs. Transformer-based models are particularly attractive because self-attention can capture global dependencies among heterogeneous features without relying solely on local or sequential processing. Korolev and Protsenko [14] demonstrated that Transformer language models can use human-readable representations containing chemical composition, crystal symmetry, and site geometry to achieve accurate and interpretable material-property predictions. Their results illustrate the potential of Transformer-based representation learning for connecting complex material descriptors with predictive outputs.

A complementary approach was proposed by Yan et al. [15], who developed a periodic graph Transformer for crystal-material property prediction. Their Matformer architecture explicitly represented periodic crystal structures and demonstrated that Transformer-based models can capture complex structural relationships that are difficult to describe through conventional fixed descriptors alone. These developments support the use of self-attention for modelling global interactions among compositional, structural, and physical features.

At the same time, the limitations of purely data-driven modelling have stimulated substantial interest in physics-informed ML. Karniadakis et al. [16] established a general framework in which physical relationships, governing equations, and domain constraints can be incorporated into learning systems. Physics-informed models are particularly useful when available data are limited or when physically inconsistent predictions must be avoided. Although much of the early physics-informed literature focused on differential-equation-based systems, the underlying principle is increasingly relevant to materials engineering, where physical descriptors and constraints can guide the learning process.

In HEA research, computational simulations and experiments are also frequently combined to resolve composition–structure–property relationships across different length scales. Jiang et al. [17] reviewed the synergistic use of first-principles calculations, molecular dynamics, phase-diagram calculations, finite-element methods, and experimental characterization for HEAs. These approaches provide physically detailed insight into phase formation, deformation, stress–strain behaviour, and thermodynamic stability, but their computational cost can become substantial when large compositional spaces are explored. This limitation creates a strong motivation for hybrid models that retain physically meaningful information while reducing the computational burden of repeated high-fidelity simulation.

Interpretability represents another important requirement for intelligent materials modelling. Swateelagna et al. [18] applied explainable ML to refractory HEA design and used shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME) analyses to identify the material parameters that contributed most strongly to phase predictions. Their results demonstrated that predictive accuracy alone is insufficient for engineering materials design; understanding which physical variables influence model outputs is also important when models are used to support scientific interpretation or alloy development.

Recent work has therefore increasingly emphasized descriptor design and physics-informed integration. Rumman et al. [19] proposed a ML-based framework to accelerate the design of HEAs. By leveraging an experimentally validated HEA dataset and incorporating 49 elemental descriptors, they developed a robust classification model and further validated phase stability through first-principles calculations.

This shift toward physics-guided modelling has also become evident in recent reviews of HEA engineering systems. Xu et al. [20] summarized phase prediction approaches based on purely ML frameworks and hybrid

physics-informed models, critically examined the challenges and limitations of current methodologies, and proposed future directions aimed at constructing interpretable models, achieving mechanistic understanding, and enabling efficient exploration of the HEA design space.

Despite these advances, several research gaps remain. First, physical knowledge and data-driven representation learning are still frequently treated as separate modelling strategies rather than integrated components of a unified architecture. Second, many HEA studies focus on individual properties or phase classification rather than simultaneously modelling mechanical response and failure behaviour. Third, the interactions among thermodynamic, compositional, microstructural, and mechanical descriptors are not always represented explicitly. Finally, model interpretability and physical consistency remain important concerns when increasingly complex deep-learning architectures are applied to engineering materials.

The present study addresses these gaps through a physics-guided hybrid Transformer framework for HEA mechanical-response and microstructural-failure prediction. Unlike purely data-driven architectures, the proposed approach combines physically informed descriptors with self-attention-based global feature interaction modelling and physics-guided optimization. The framework therefore provides a unified basis for prediction, interpretation, and evaluation of the relationships among material constitution, microstructure, deformation behaviour, and failure.

3 Materials and Methods

3.1 Proposed Physics-Guided Transformer Framework

A physics-guided hybrid Transformer modelling framework was developed to predict the mechanical response and microstructural failure behaviour of HEAs. The framework integrates physically meaningful descriptors derived from materials science with Transformer-based representation learning, allowing compositional, thermodynamic, microstructural, and mechanical information to be analysed within a unified modelling architecture. The objective of this integration is to establish nonlinear relationships among alloy composition, microstructural characteristics, and failure-related mechanical properties while retaining physically interpretable information in the learning process.

Unlike conventional ML approaches that rely primarily on statistical correlations, the proposed framework incorporates material-specific physical information associated with phase stability, lattice distortion, and thermodynamic behaviour. These physical descriptors are introduced both at the feature-representation stage and through physics-guided regularization during model optimization. The resulting architecture therefore combines domain-informed material representation with data-driven attention learning rather than treating the two components independently. This hybrid modelling strategy is intended to improve predictive reliability, generalization, and interpretability when analysing compositionally and microstructurally complex HEA systems.

The complete modelling procedure consists of the following stages:

1. Acquisition and preprocessing of HEA data;
2. Physics-guided feature extraction;
3. Transformer-based representation learning;
4. Attention-based analysis of interactions among material descriptors;
5. Mechanical property prediction;
6. Microstructural failure-probability estimation; and
7. Model optimization, validation, and performance evaluation.

The overall architecture of the proposed framework is illustrated in Figure 1.

3.2 Dataset Preparation and Normalization

The HEA dataset consisted of experimentally reported alloy systems containing FCC, BCC, FCC+BCC, intermetallic, and amorphous microstructural phases. The dataset included compositional, microstructural, and mechanical information, including grain size, hardness, density, UTS, and failure strain. These variables were used together with physics-guided descriptors to construct the input representation for subsequent Transformer learning.

Before model training, all numerical features were rescaled using min–max normalization:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

where, X represents the original feature value, and $X_{\min}$ and $X_{\max}$ denote the minimum and maximum feature values, respectively.

Normalization mapped features with different numerical ranges onto a comparable scale, thereby reducing scale-induced imbalance during optimization and improving numerical stability and convergence during model training.

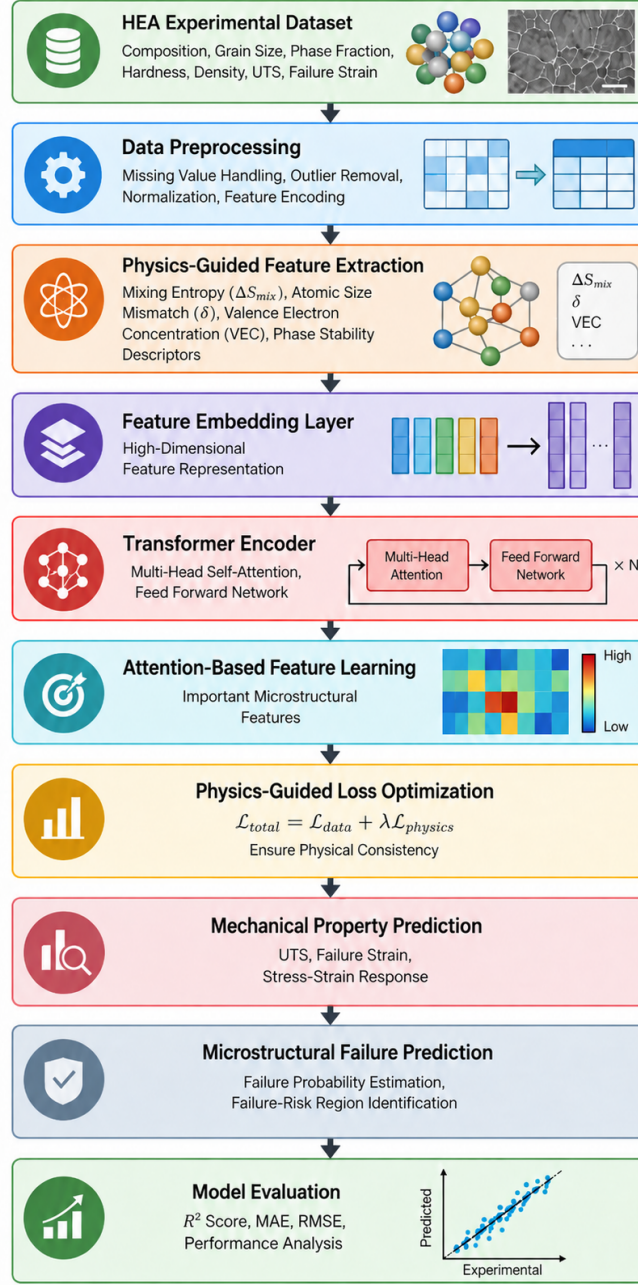


Figure 1. Overall architecture of the proposed physics-guided hybrid Transformer framework for high-entropy alloys (HEAs) microstructural failure and mechanical response prediction

Note: UTS: ultimate tensile strength; MAE: mean absolute error; RMSE: root mean square error.

3.3 Physics-Guided Feature Extraction

To incorporate domain knowledge into the data-driven learning process, the framework included thermodynamic and material descriptors commonly used to characterize the phase formation, lattice distortion, and stability of HEAs. These descriptors were not treated merely as additional statistical inputs; rather, they provided physically meaningful information linking alloy composition to microstructural and mechanical behaviour.

Mixing Entropy

The configurational mixing entropy was calculated as follows:

$$\Delta S_{\text{mix}} = -R \sum_{i=1}^n c_i \ln c_i \quad (2)$$

where, R is the gas constant, and c_i represents the atomic concentration of the corresponding alloying element.

Mixing entropy describes the configurational contribution arising from the multi-component nature of HEAs and is closely related to phase formation and thermodynamic stability.

Atomic Size Mismatch

The atomic size mismatch was calculated as follows:

$$\delta = \sqrt{\sum_{i=1}^n c_i \left(1 - \frac{r_i}{\bar{r}}\right)^2} \quad (3)$$

where, r_i denotes the atomic radius of the corresponding element, and $\bar{r}$ represents the average atomic radius of the alloy system.

This descriptor characterizes lattice distortion induced by differences in atomic size among constituent elements and therefore provides information relevant to strengthening, phase stability, and deformation behaviour.

Valence Electron Concentration

The valence electron concentration was calculated as follows:

$$\text{VEC} = \sum_{i=1}^n c_i (\text{VEC})_i \quad (4)$$

The valence electron concentration provides an additional descriptor of electronic structure and phase stability in HEAs. Together with mixing entropy, atomic size mismatch, and the other material descriptors used in the framework, it contributes physically informed information that complements the experimentally reported microstructural and mechanical variables.

The complete physics-guided feature set was subsequently combined with the normalized experimental features and used as the input to the Transformer-based representation-learning module. In this way, the proposed framework integrated domain-informed material characteristics with data-driven nonlinear feature interaction modelling.

3.4 Transformer-Based Representation Learning

The data-driven component of the proposed framework was constructed using a Transformer encoder architecture. The Transformer was selected because its self-attention mechanism can directly represent nonlinear and long-range interactions among heterogeneous material descriptors, which is particularly important when composition, thermodynamic parameters, phase information, microstructural characteristics, and mechanical properties interact simultaneously.

The input feature vector was defined as follows:

$$X = [x_1, x_2, x_3, \dots, x_n] \quad (5)$$

where the feature vector contains the normalized experimental variables together with the physics-guided descriptors introduced in Section 3.3.

The self-attention operation was defined as:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \quad (6)$$

where, Q is the query matrix, K is the key matrix, V is the value matrix, and d_k is the dimensional scaling factor.

The attention mechanism determines the relative dependence among input descriptors and enables the model to assign different levels of importance to compositional, thermodynamic, microstructural, and mechanical information. Multi-head attention was employed to capture different interaction patterns within the feature space. Through this mechanism, the model learned global relationships among phase structure, grain characteristics, alloy descriptors, and mechanical behaviour rather than relying exclusively on local or pairwise correlations.

The Transformer output therefore represents a learned latent description of the HEA system in which physics-guided information and data-driven feature interactions are jointly encoded for subsequent mechanical-property and failure prediction.

3.5 Physics-Guided Loss Optimization

To further incorporate physical consistency into model training, the optimization objective combined prediction error (L_{data}) with a physics-guided regularization term ($\lambda L_{\text{physics}}$, λ is the weight coefficient). The overall loss function (L_{total}) was defined as:

$$L_{\text{total}} = L_{\text{data}} + \lambda L_{\text{physics}} \quad (7)$$

The prediction-loss component was calculated as:

$$L_{\text{data}} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (8)$$

where, N denotes the number of samples, y_i denotes the actual mechanical property, and $\hat{y}_i$ denotes the corresponding predicted value.

The prediction term minimizes the discrepancy between model outputs and experimental observations, whereas the physics-guided term constrains the optimization process toward material behaviour that remains consistent with the physical descriptors and relationships incorporated into the framework.

This combined objective establishes a second level of integration between physics-based knowledge and data-driven learning. Physical information is therefore introduced not only through the input representation but also through the optimization process. The resulting model is designed to reduce physically inconsistent predictions while maintaining the nonlinear representation capability of Transformer learning.

3.6 Mechanical Property and Failure Prediction

The learned latent representation was used to predict multiple aspects of mechanical and failure behaviour. The proposed framework considered the following outputs:

- UTS;
- failure strain;
- stress–strain response; and
- microstructural failure probability.

For continuous mechanical-property prediction, the learned representation was mapped to the corresponding regression outputs. Failure probability was estimated using a sigmoid activation function:

$$P_f = \frac{1}{1 + e^{-z}} \quad (9)$$

where, P_f represents the failure probability, and z denotes the latent feature representation generated by the model.

The sigmoid function maps the model output onto a probability range between 0 and 1. A higher predicted value corresponds to a greater estimated likelihood of microstructural failure. This probabilistic output complements the deterministic prediction of tensile and deformation properties and enables the framework to characterize both mechanical response and failure risk within the same modelling structure.

The integration of these outputs is particularly important for HEA assessment because high tensile strength alone does not fully describe failure behaviour. Simultaneous prediction of strength, strain, stress–strain response, and failure probability provides a more comprehensive representation of mechanical performance.

3.7 Model Optimization and Evaluation

The proposed model was trained using the Adam optimization algorithm:

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} \quad (10)$$

where, θ is the model parameter, η is the learning rate, $\hat{m}_t$ is the bias-corrected first moment estimate, $\hat{v}_t$ is the bias-corrected second raw moment estimate, ϵ is the smoothing term, and t is the time step.

The available dataset was divided into:

- 80% training data; and
- 20% testing data.

The training subset was used to optimize model parameters, whereas the testing subset was retained for independent performance evaluation.

Predictive performance was assessed using the following metrics:

- coefficient of determination (R^2);
- mean absolute error (MAE); and
- root mean square error (RMSE).

The evaluation metrics were defined as follows:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (11)$$

$$\mathbf{MAE} = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (12)$$

$$\mathbf{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (13)$$

R^2 measures the proportion of variation in the observed values explained by the model, whereas MAE quantifies the average absolute prediction error and RMSE gives greater weight to larger prediction deviations. The combined use of these metrics enables the framework to be evaluated from complementary perspectives of explanatory capability, average prediction error, and sensitivity to large deviations.

In addition to quantitative prediction metrics, the modelling framework was evaluated through comparisons of predicted and observed mechanical behaviour, stress–strain response analysis, failure-probability assessment, attention-based feature interpretation, and ablation analysis. These evaluations were designed to determine not only whether the framework achieves accurate prediction but also whether the integration of physics-guided descriptors, Transformer attention, and physics-guided optimization contributes meaningfully to model performance and interpretability.

4 Results and Analysis

4.1 Dataset Distribution Analysis

The distribution of HEA samples across different microstructural phase categories is presented in Figure 2a. The dataset comprises FCC, BCC, FCC+BCC, intermetallic, and amorphous phase categories, with FCC, BCC, and mixed FCC+BCC systems representing the largest groups. This phase diversity is important for the present modelling task because phase constitution is closely associated with the mechanical strength, ductility, and failure behaviour of HEAs.

The inclusion of multiple phase categories also provides the proposed physics-guided hybrid Transformer framework with heterogeneous material configurations from which composition–microstructure–property relationships can be learned. In particular, differences in phase constitution, grain characteristics, and thermodynamic descriptors introduce variations in mechanical response that must be represented by the model when predicting strength, deformation, and failure-related properties. The phase-wise distribution therefore provides an appropriate basis for evaluating the framework across HEAs with different microstructural characteristics.

4.2 Training and Validation Loss Analysis

The training and validation loss curves of the proposed physics-guided hybrid Transformer model are shown in Figure 2b. Both losses decreased progressively during training and approached stable levels as the optimization proceeded, indicating effective convergence of the learning process. The relatively small separation between the training and validation curves further suggests that substantial overfitting did not occur during model development.

The validation loss followed the overall trend of the training loss rather than increasing as training progressed. This behaviour indicates that the learned representation retained predictive capability for samples not directly used for parameter optimization. Within the proposed framework, this convergence reflects the joint optimization of data-driven Transformer representation learning and the physics-guided components incorporated into the modelling process. The specific contribution of these components to predictive performance is examined further through the comparative and ablation analyses presented in Sections 4.5 and 4.9.

4.3 Ultimate Tensile Strength Prediction

The relationship between predicted and experimentally measured UTS is presented in Figure 2c. Most predicted values were distributed close to the diagonal reference line, indicating good agreement between model estimates and experimental observations over the represented strength range.

This agreement shows that the proposed framework captured relevant relationships among alloy composition, physics-guided descriptors, microstructural characteristics, and tensile strength. The result is particularly relevant to the present modelling objective because UTS represents a key engineering indicator of load-bearing capacity and failure resistance in HEAs. Accurate prediction of this property therefore provides an important basis for evaluating whether the hybrid representation can translate heterogeneous material information into mechanically meaningful outputs.

4.4 Failure Strain Prediction

Figure 2d compares the predicted and experimentally observed failure strain values. The predictions were generally distributed close to the corresponding measured values across the represented strain range, indicating that the framework captured the principal variation in ductility and deformation capacity within the dataset.

Failure strain describes the deformation sustained before fracture and therefore complements UTS in characterizing the mechanical failure behaviour of HEAs. Whereas UTS primarily reflects the maximum tensile load that the material can sustain, failure strain provides information on deformation capacity before fracture. The agreement observed in Figure 2d consequently supports the ability of the proposed framework to represent both strength- and deformation-related aspects of material failure rather than relying on a single mechanical indicator.

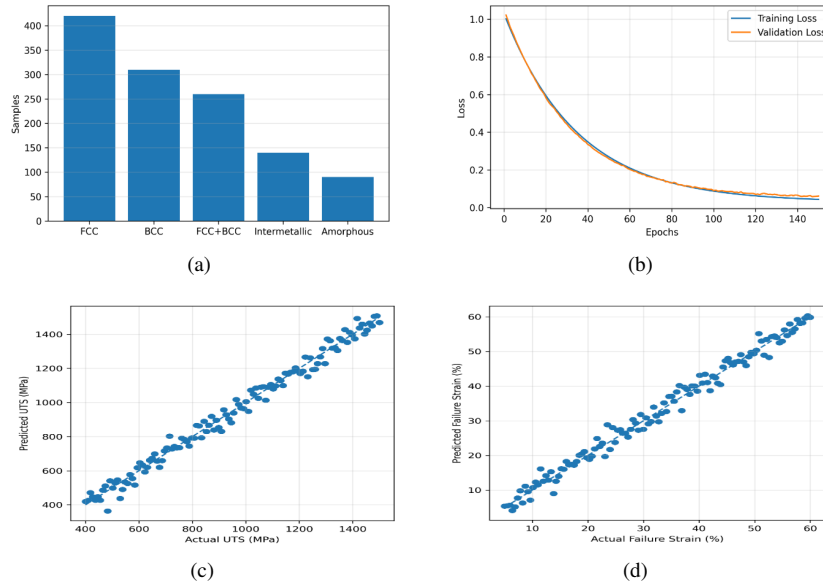


Figure 2. (a) Distribution of high-entropy alloy (HEA) samples by microstructural phase category. (b) Training and validation loss curves of the proposed physics-guided hybrid Transformer framework. (c) Predicted versus measured ultimate tensile strength (UTS). (d) Predicted versus measured failure strain

Note: FCC: face-centered cubic; BCC: body-centered cubic.

4.5 Model Performance Comparison

The predictive performance of random forest, extreme gradient boosting (XGBoost), convolutional neural network (CNN), long short-term memory (LSTM), the standard Transformer, and the proposed physics-guided hybrid Transformer framework is compared in Figure 3a. Among the evaluated models, the proposed framework achieved the highest R^2 score.

The difference relative to random forest and XGBoost indicates the benefit of learning more complex nonlinear interactions among heterogeneous material descriptors. The comparison with CNN and LSTM further shows the advantage of a representation architecture capable of modelling broader interactions among compositional, thermodynamic, microstructural, and mechanical features. Importantly, the proposed framework also outperformed the standard Transformer, indicating that the performance improvement cannot be attributed solely to the use of self-attention.

Instead, the comparison suggests that integrating physics-guided material information with Transformer-based representation learning contributed additional predictive value. This distinction is central to the proposed hybrid modelling strategy: the Transformer provides data-driven representation of complex feature interactions, whereas the physics-guided components introduce material-domain information into the modelling process. The individual contribution of these components is examined more directly in the ablation study presented in Section 4.9.

4.6 Stress–Strain Behavior Analysis

The experimental and predicted stress–strain curves are compared in Figure 3b. The predicted response followed the overall experimental trend across both the elastic and plastic deformation regions. The agreement indicates that the proposed framework reproduced the principal evolution of mechanical response rather than only estimating isolated scalar properties such as UTS and failure strain.

This result is important because mechanical failure develops through an evolving deformation process. Reproducing the stress–strain response therefore provides a more comprehensive assessment of model performance than point-wise property prediction alone. The observed correspondence between the predicted and experimental curves supports the use of the physics-guided hybrid representation for characterizing deformation behaviour and provides additional evidence of its potential for failure analysis and mechanical reliability assessment of HEAs.

4.7 Microstructural Failure-Probability Analysis

The microstructural failure-probability heatmap generated by the proposed model is presented in Figure 3c. Variations in heatmap intensity represent differences in the estimated probability of failure, with higher-intensity regions corresponding to higher predicted failure probability and lower-intensity regions indicating comparatively lower predicted failure probability.

The probability distribution provides a visual representation of how the model differentiates failure risk across the analysed feature space. In conjunction with information on phase constitution, grain characteristics, and other microstructural descriptors, the heatmap facilitates interpretation of the conditions associated with elevated predicted failure risk. This probabilistic representation complements the deterministic predictions of UTS, failure strain, and stress–strain response and extends the framework from mechanical-property estimation to failure-oriented assessment.

The heatmap should therefore be interpreted as a model-based representation of predicted failure probability rather than direct experimental imaging of local failure sites. Within this interpretation, it provides an additional means of examining the relationship between microstructural characteristics and predicted failure behaviour and can support subsequent microstructure-oriented material assessment.

4.8 Transformer Feature Attention Analysis

The Transformer-based feature attention results are shown in Figure 3d. The attention distribution assigned comparatively high importance to grain size and phase fraction, followed by valence electron concentration, entropy, hardness, density, and heat-treatment temperature. This pattern indicates that the model relied on a combination of microstructural, thermodynamic, and mechanical descriptors when constructing its representation of HEA failure behaviour.

The relatively high attention assigned to grain size and phase fraction is physically meaningful because these characteristics are closely associated with deformation and strengthening behaviour, phase stability, dislocation activity, and fracture resistance. At the same time, the contributions of valence electron concentration and entropy indicate that the learned representation also incorporated physics-guided descriptors rather than depending exclusively on experimentally measured mechanical variables.

The attention analysis improves model transparency by identifying the features that received greater emphasis during representation learning. However, attention magnitude is interpreted here as an indicator of model feature relevance rather than direct evidence of physical causality. Within this limitation, the observed attention pattern provides a useful connection between the data-driven Transformer representation and physically meaningful material characteristics.

4.9 Ablation Study

The contribution of the principal modelling components was further examined through the ablation study presented in Figure 4. Performance was compared across the Transformer-only model, the Transformer with attention, the Physics-Guided Transformer, and the complete proposed framework. The complete framework achieved the highest R^2 , higher failure-prediction accuracy, and the lowest UTS mean absolute error among the evaluated configurations.

The progressive differences among the ablated configurations indicate that the components of the proposed architecture contributed differently to overall model performance. The Transformer component provided the basic representation-learning capability for nonlinear relationships among material features, while the attention mechanism improved the model’s ability to differentiate the relevance of individual descriptors. Incorporating physics-guided information further improved the consistency and predictive representation of material behaviour.

The best overall performance was obtained when these components were combined in the complete framework. This result supports the central premise of the proposed hybrid modelling architecture: predictive performance benefits from the structured integration of data-driven representation learning, attention-based feature interaction modelling, and physics-guided material information rather than from any individual component alone. The ablation analysis therefore provides direct evidence for the contribution of the hybrid integration strategy to prediction accuracy, interpretability, and physical consistency.

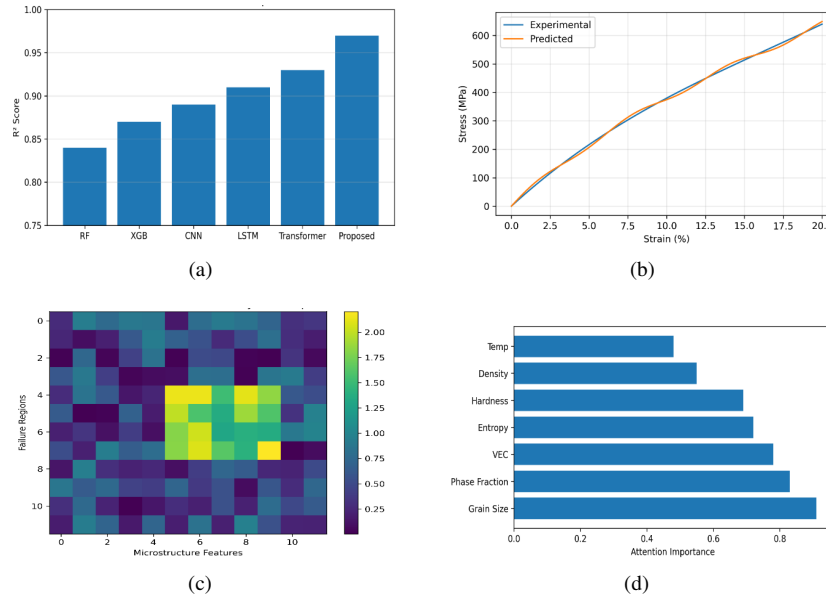


Figure 3. (a) Comparison of model performance based on R^2 . (b) Experimental and predicted stress–strain responses. (c) Microstructural failure-probability heatmap generated by the proposed framework. (d) Transformer-based attention analysis of features associated with high-entropy alloy (HEA) failure prediction. Note: RF: random forest; XGB: extreme gradient boosting; CNN: convolutional neural network; LSTM: long short-term memory; Temp: temperature; VEC: valence electron concentration.

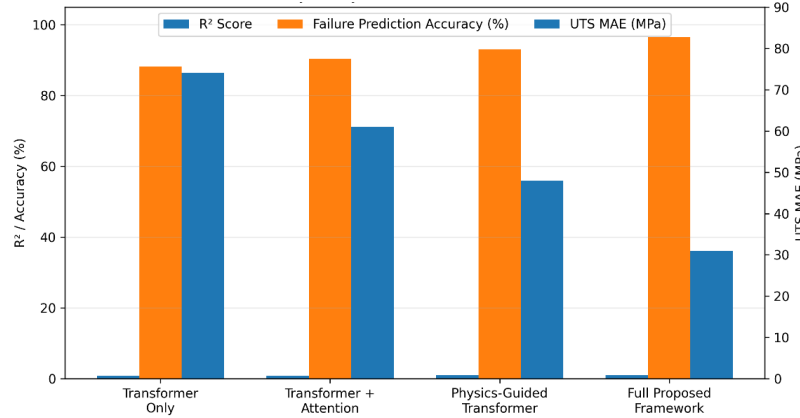


Figure 4. Ablation analysis of the principal components of the proposed physics-guided hybrid Transformer framework

Note: UTS MAE: mean absolute error of ultimate tensile strength.

5 Discussion

The results demonstrate the potential of the proposed physics-guided hybrid Transformer modelling framework for predicting mechanical response and microstructural failure behaviour in HEAs. The principal advantage of the framework lies not simply in the use of a Transformer architecture, but in the structured integration of physically meaningful material descriptors with data-driven representation learning. By incorporating information associated with thermodynamic behaviour, lattice distortion, phase stability, and microstructural characteristics, the framework establishes a direct connection between material-domain knowledge and the nonlinear feature interactions learned through self-attention.

The comparison with random forest, XGBoost, CNN, LSTM, and the standard Transformer provides insight into the contribution of this hybrid modelling strategy. The higher predictive performance of the proposed framework relative to conventional ML models indicates that nonlinear interactions among heterogeneous material descriptors cannot be fully represented by statistical learning alone. Its improvement over CNN and LSTM further suggests that global interactions among compositional, thermodynamic, and microstructural variables are important for HEA property prediction. More importantly, the improvement over the standard Transformer indicates that self-attention

alone does not account for the performance of the complete framework. The incorporation of physics-guided information provides an additional source of material-specific structure within the learning process.

The ablation results further support this interpretation by showing that the complete framework achieved better overall performance when its physics-guided and data-driven components were integrated. This finding is important from a hybrid modelling perspective because the individual components serve different but complementary functions. Physics-guided descriptors introduce prior knowledge of material behaviour, Transformer representation learning captures nonlinear interactions among heterogeneous features, and attention provides information on the relative relevance of these features within the learned representation. Their integration therefore provides a more structured approach to HEA failure prediction than either purely descriptor-based ML or unconstrained deep learning.

The stress–strain analysis extends this assessment beyond the prediction of individual scalar properties. The correspondence between predicted and experimental stress–strain responses indicates that the framework captured the principal evolution of mechanical deformation in addition to UTS and failure strain. This capability is relevant to engineering assessment because failure cannot be characterized adequately by a single strength indicator. The combined prediction of strength, deformation capacity, stress–strain response, and failure probability provides a broader representation of mechanical behaviour and allows the model to address different aspects of material reliability within a unified framework.

Interpretability is another important aspect of the proposed approach. The attention analysis assigned comparatively high importance to grain size and phase fraction, followed by valence electron concentration, entropy, hardness, density, and heat-treatment temperature. The prominence of both conventional microstructural variables and physics-guided descriptors indicates that the learned representation draws on multiple sources of material information. Grain size and phase fraction are particularly relevant to deformation, strengthening, and fracture behaviour, while valence electron concentration and entropy provide complementary information associated with the physical characteristics of multi-component alloy systems. Attention weights should not be interpreted as direct evidence of causal mechanisms; nevertheless, they provide a useful indication of the variables emphasized by the model and improve the transparency of the prediction process.

Similarly, the failure-probability heatmap provides a model-based representation of variations in predicted failure risk. Rather than constituting direct experimental evidence of spatial failure initiation, it should be interpreted as a visualization of the failure probabilities estimated from the learned feature representation. Used together with attention analysis and mechanical-property predictions, this representation provides an additional means of examining how the model associates heterogeneous material characteristics with failure-related outcomes.

From an engineering modelling perspective, these findings suggest that physics-guided and data-driven approaches need not be treated as competing alternatives. Their integration can combine the representation capability of modern learning architectures with material-domain information that constrains and contextualizes the prediction process. This is particularly relevant to HEAs, for which large compositional spaces and strongly coupled microstructural variables make exhaustive experimental exploration difficult. The proposed framework therefore provides a potentially useful modelling structure for data-assisted alloy assessment, failure-risk analysis, and microstructure–property mapping.

Several limitations should nevertheless be recognized. The performance of the framework remains dependent on the coverage, quality, and representativeness of the available experimental HEA data. The present feature representation is predominantly descriptor-based and does not explicitly incorporate spatial microstructural information obtained directly from microscopy. Moreover, the current validation is limited to the material systems and loading conditions represented in the available dataset. Further evaluation across broader alloy compositions, processing histories, loading regimes, and thermal conditions is therefore required before the framework can be considered broadly transferable. Future development could integrate microscopy-derived image features, finite-element simulation outputs, and additional experimental observations within the hybrid architecture. Such extensions would allow the framework to combine multiple sources of physical and data-driven information and provide a more comprehensive representation of microstructure-dependent failure behaviour.

6 Conclusion

This study developed a physics-guided hybrid Transformer modelling framework for predicting mechanical response and microstructural failure behaviour in HEAs. The framework integrated physically meaningful descriptors with Transformer-based representation learning to capture relationships among alloy composition, microstructural characteristics, thermodynamic information, and mechanical behaviour. Mixing entropy, atomic size mismatch, valence electron concentration, phase-related information, and microstructural features provided the physics-guided representation, while self-attention enabled nonlinear interactions among heterogeneous material descriptors to be learned within a unified modelling architecture.

The results showed close agreement between predicted and experimentally measured UTS and failure strain, while the predicted stress–strain response reproduced the principal deformation trend observed in the experimental data.

The proposed framework also achieved better predictive performance than random forest, XGBoost, CNN, LSTM, and the standard Transformer in the reported comparisons. The ablation analysis further showed that integrating the principal physics-guided and data-driven components improved overall model performance relative to the evaluated reduced configurations.

The failure-probability analysis and Transformer attention results provided complementary information for interpreting the model outputs. Grain size, phase fraction, valence electron concentration, entropy, hardness, density, and heat-treatment temperature received different levels of attention within the learned representation, demonstrating that both microstructural variables and physics-guided descriptors contributed to the prediction process. These interpretability analyses do not establish causal failure mechanisms, but they improve transparency regarding the material characteristics emphasized by the model.

Overall, the findings demonstrate that the structured integration of physics-guided material information and Transformer-based representation learning provides an effective hybrid modelling strategy for HEA mechanical-property and failure prediction. The framework offers a physically informed and interpretable basis for microstructure–property analysis, failure-risk assessment, and data-assisted alloy design. Future work should extend the available experimental dataset and incorporate microscopy-derived microstructural information, finite-element simulation data, and observations obtained under broader loading and thermal conditions. Such developments could further improve the robustness, transferability, and engineering applicability of hybrid intelligent modelling for complex alloy systems.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflicts of interest.

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